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SHENZHEN SILICONTRA TECHNOLOGY CO.,LTD.



PAN3029ATR4-GC

433MHz Wireless Transceiver Module User Specification

(V1.0)

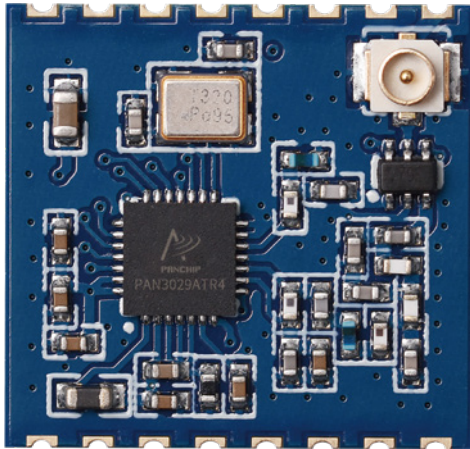
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Document Revision History

Version	Change date	Change Notes
V1.0	October 23, 2024	Initial version

1 Module Introduction



(Front of the module)



(Back of the module)

(The actual module shall prevail)

1.1 Module Overview

The PAN3029ATR4-GC is a module independently developed and designed by our company based on the PAN3029 RF chip from the domestic manufacturer, Pantech Microelectronics.

The biggest highlight of the PAN3029ATR4-GC module is its transmit power reaching +20dBm and receive sensitivity reaching -143dBm. Combined with its advanced ChirpLoTTM modulation and demodulation technology, it significantly improves communication range. It holds a strong position in short-range IoT wireless communication applications.

The PAN3029ATR4-GC module features small size, low power consumption, long transmission distance, and strong anti-interference capabilities, making it widely applicable in various wireless communication fields within the IoT ecosystem.。

1.2 Module Features

- Supports 433MHz band with strong penetration
- Maximum transmit power 20dBm, software-adjustable receive current 4.1mA@DCDC mode
- SPI communication interface, directly connectable to various microcontrollers, wide operating voltage range 1.8~3.6V
- Industrial-grade standard design, supports long-term use at -40~+85°C, ultra-small size, only 16x17mm
- Stamp-perforation design for easy mass production

1.3 Application Scenarios

- Home security alarms
- Healthcare
- Smart home systems
- Building Automated Data Collection Systems
- Smart parking systems
- Automotive Industry Applications
- Agricultural automation solutions
- Wireless industrial-grade remote controls, industrial sensors, and other products

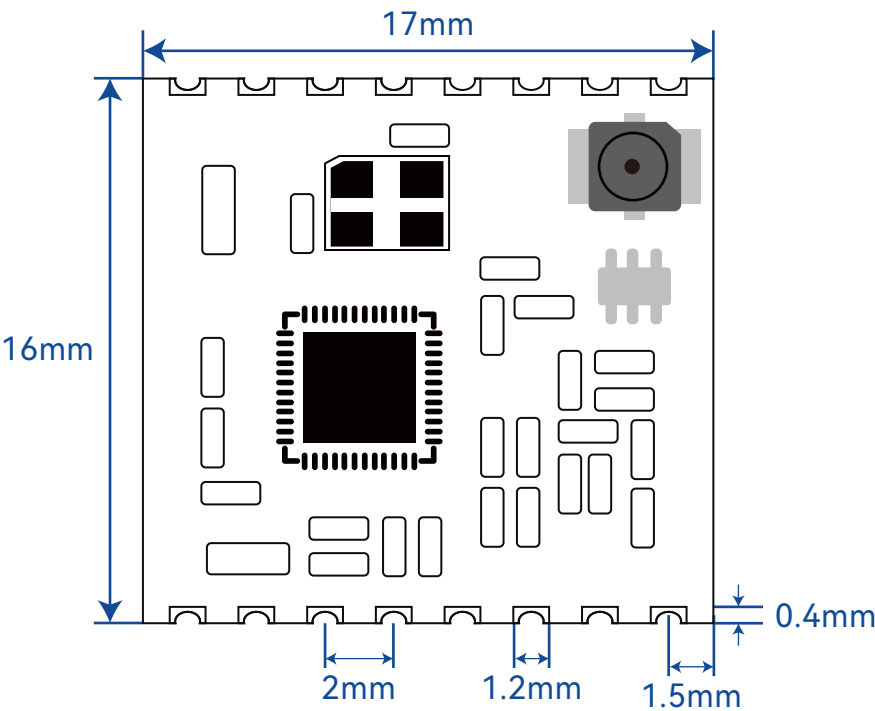
2 Module Parameters

2.1 Basic Electrical Parameter Diagram of the Module

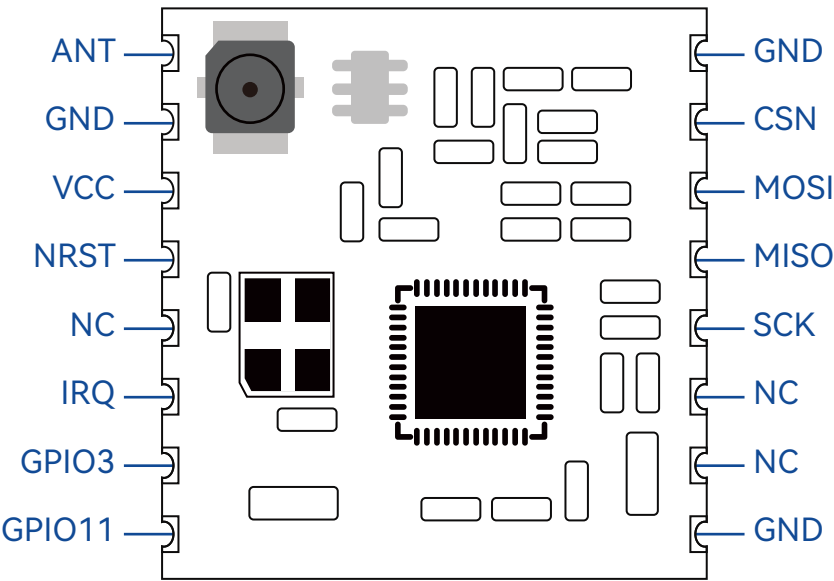
Project	Specification		Description
Parameters	Characteristic		Note
Operating voltage	1.8 ~ 3.6V		Typical 3.3V
Operating temperature	-40 ~ 85℃		
Operating frequency	408 ~ 550MHz		433MHz is recommended; software configurable
Power consumption	Transmitting	96mA@20dBm	At maximum transmit power
	Receiving	8mA@LDO / 4.1mA@DC-DC	
	Sleep	260nA	
Transmit power	20dBm		Maximum value, user-programmable
Receiver sensitivity	-143dBm		@62.5KHz
Modulation method	ChirpIoT™		
Modulation parameters	BW : 62.5KHZ~500KHZ ; SF : 5~12 ; CR : 4/5, 4/6, 4/7, 4/8		
Mounting	Stamp holes		
Communication interface	SPI		
Dimensions	16*17mm		
Antenna type	External antenna for stamp holes, IPEX mounted		Equivalent impedance approximately 50Ω

3 Module Description

3.1 Module Dimension Diagram



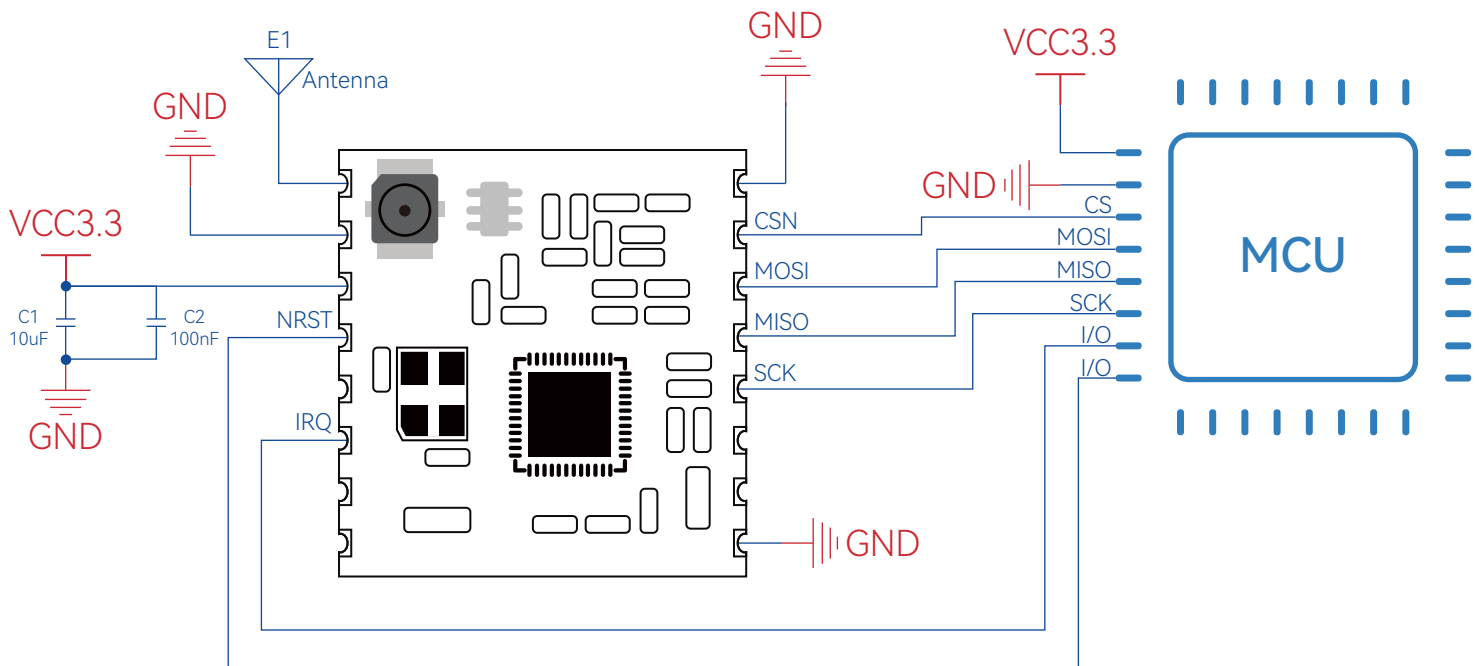
3.2 Module Pin Function Definition Diagram



3.3 Pin Function Description

Pin No.	Name	Function Description
1	ANT	Antenna interface, equivalent impedance approximately 50Ω
2	GND	Ground
3	VCC	Module power supply pin, nominal supply voltage 3.3V
4	NRST	Reset pin - active low
5	NC	Not connected
6	IRQ	Interrupt signal
7	GPIO3	Digital I/O
8	GPIO11	Digital I/O or channel status indicator signal
9	GND	Ground
10	NC	Not connected
11	NC	Not connected
12	SCK	SPI bus clock pin, normally at a high level
13	MISO	SPI bus slave output pin
14	MOSI	SPI bus slave input pin
15	CSN	SPI chip select pin, active low
16	GND	Ground

3.4 Module Connection Diagram



4 Additional Notes

1. It is recommended to use a linear regulated power supply (LDO) to power this module. The power supply ripple should be as low as possible, and the module must be reliably grounded. Please ensure the correct connection of the positive and negative terminals of the power supply. Reversing the connection may cause permanent damage to the module.
2. Do not surround the module antenna with other metal objects, otherwise it will seriously affect the communication distance.

5 Antenna Selection

5.1 Antenna Usage Precautions

- The antenna mounting structure significantly impacts module performance. For better performance, the antenna should be exposed, ideally vertically upwards. When the module is installed inside the housing, a high-quality antenna extension cable can be used to extend the antenna to the outside of the housing. If the product does not allow for exposure, a spring antenna or FPC antenna must be used.
- Installing the antenna inside a metal casing will drastically reduce the transmission distance.
- If purchasing a suction cup antenna, the lead wire should be as straight as possible, and the suction cup base should be attached to a metal object whenever possible.



433 MHz Spring Antenna



433 MHz FPC Antenna



433 MHz Rod Antenna



433 MHz Suction Cup Antenna

6 Hardware Design

- It is recommended to use a DC regulated power supply to power this module. The power supply ripple factor should be minimized, and the module must be reliably grounded.
- Please ensure the correct connection of the positive and negative terminals of the power supply. Reversed connection may cause permanent damage to the module.
- Please check the power supply and ensure it is within the recommended supply voltage range. Exceeding the maximum value will cause permanent damage to the module.
- Please check the power supply stability; the voltage should not fluctuate significantly or frequently.
- When designing the power supply circuit for the module, it is generally recommended to retain a margin of at least 30% to facilitate long-term stable operation of the entire device.
- The module should be kept as far away as possible from power supplies, transformers, high-frequency traces, and other areas with high electromagnetic interference.
- High-frequency digital traces, high-frequency analog traces, and power supply traces must avoid running under the module. If it is absolutely necessary to run them under the module, assuming the module is soldered on the Top Layer, a ground copper layer should be laid on the Top Layer at the module contact area (all copper and well grounded), close to the digital section of the module and routed on the Bottom Layer.
- If the module is soldered or placed on the Top Layer, arbitrarily routing traces on the Bottom Layer or other layers is also incorrect and will affect the module's spurious emissions and receiver sensitivity to varying degrees.
- If devices around the module generate significant electromagnetic interference, it will greatly affect the module's performance. Depending on the intensity of the interference, it is recommended to keep them appropriately away from the module. If possible, implement appropriate isolation and shielding.
- If traces around the module generate significant electromagnetic interference (high-frequency digital, high-frequency analog, power supply traces), it will also greatly affect the module's performance. Depending on the intensity of the interference, it is recommended to keep them appropriately away from the module. If possible, implement appropriate isolation and shielding.

7 Unsatisfactory Transmission Distance

- Communication distance will decrease when there are obstacles or obstructions in line-of-sight communication.
- Temperature, humidity, and co-channel interference can increase packet loss rate.
- Ground absorption and reflection of radio waves result in poor performance when tested near the ground.
- The presence of metal objects near the antenna, or placement within a metal enclosure, will cause severe signal attenuation. An excessively high airspeed setting (higher airspeed results in shorter distances) can also cause problems.
- Lower-than-recommended power supply voltage at room temperature reduces transmission power.
- Poor antenna-module compatibility or antenna quality issues.

8 Module Easily Damaged

- Please check the power supply and ensure it is within the recommended voltage range. Exceeding the maximum value will cause permanent damage to the module.
- Please check the power supply stability; the voltage should not fluctuate.
- Please ensure anti-static operation during installation and use, as high-frequency components are electrostatically sensitive.
- Please ensure that the humidity is not too high during installation and use, as some components are humidity-sensitive.
- Unless there are special requirements, it is not recommended to use the product at excessively high or low temperatures.

9 High Bit Error Rate

- Interference from nearby co-channel signals: Move away from the interference source or change the frequency or channel to avoid interference.
- Inadequate power supply can also cause garbled characters; ensure power supply reliability.
- Poor quality or excessively long extension cables or feeders can also lead to a higher bit error rate.